Team Control Number
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Problem Chosen
B
2023 HiMCM
Summary Sheet

Electrifying a Metro Bus Fleet Under a Preserved Service Floor

Question and baseline. We study a conditional conversion of TransLink's fixed-route bus service to zero-tailpipe electric propulsion and all-electric accessories by 2033, subject to retaining recovered fall-2023 departures and capacity proxies. Existing trolleybuses count as electric; ordinary renewal is the comparator. Electricity-source compliance is treated separately.

Model. We state the full master model, then execute bounded submodels: trip and SOC validation, a route-level duty sample, a proportional city inventory, and age- or capacity-constrained cohort models. Service cannot be converted before a facility-readiness wave. Gross and grant-adjusted costs use constant 2023 CAD; emissions use the same annual service ledger and explicit partial boundaries. A city-wide Pareto frontier remains unavailable.

Observed result. The recovered fall-2023 GTFS contains 54,497 trips in 25 service patterns. The representative weekday has 1,752 source blocks and 1,087 simultaneous trips; the maximum trip day has 22,494 trips. Annual fixed-route service is 5.58 million hours and 106.87 million km. A fuel- and stock-weighted tailpipe proxy gives 126–408 tonnes NO_x per year. Vehicle identity, age, depot and firm power remain unresolved.[2, 14, 43, 44, 45]

Conditional inventory and cost scenario. At 300 km, the representative weekday yields 1,698–1,741 distance requests; all 25 recovered patterns reach at most 1,745. An assumed class inventory buys 1,647 route vehicles and 262 common trolley renewals (1,909 electric purchases), ending at 1,927 electric and zero conventional stock. Central constant-price PV is CAD 4.43 billion versus CAD 2.62 billion (-68.8%); symmetric 0/25/50% grants give $-68.8/-38.8/-5.5\%$. With retained-asset and infrastructure terminal values the score spans -41.1 to -106.4%. The partial environmental boundary spans -14.3 to 16.2%. These are inventory scenarios, not dispatch certificates. London's 17-year renewal peak is 776 slots per year.[10, 18, 14, 24, 26]

Recommendation. Start depot power and grid work in 2024, publish each site's release date, and procure against commissioned capacity. Validate a terminal-level duty plan before treating block ratios as fleet counts. Gate double-deck and shuttle conversion on in-route charging, winter tests and verified payload.

Keywords: electric bus; fleet replacement; service preservation; charging; life-cycle cost.

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1 Problem definition and study boundary

The competition asks for environmental and financial consequences, a ten-year transition in three metropolitan areas, and a decision-maker letter.[1] We take end-2023 as Vancouver's decision baseline and plan 2024–2033. Observed fall schedules and annual service totals define the service evidence; 2022 and 2025 fleet snapshots only seed named stock scenarios. No later purchase is treated as a known 2023 decision. Metro Vancouver had 2,642,825 residents in 2021, above the problem threshold.[5] Chicago and London use their separately dated 2023 registers; London's March audit is not treated as December stock.

Prior fleet models optimize electric vehicle count and mix jointly with charging and scheduling,[35] use a multi-period purchase, salvage and demand-charge formulation,[36] or solve multi-depot time-window and recharge constraints.[37] This study connects those ideas to a recovered public timetable, a capacity proxy where the vehicle roster is absent, a normal-renewal comparator, and

a separately stated emissions boundary. The result is also a data-validity test: a missing garage or winter record changes the status to `UNVERIFIED` rather than becoming a zero-cost assumption.

We adopt an operational endpoint: electric propulsion, all-electric accessories and zero tailpipe combustion. The problem's phrase "zero-emission electric source" can also require a verified supply chain. No contract, hourly matching or certificate is in evidence, so the strict-source endpoint remains `UNVERIFIED`; the environment model includes an average-grid attribution and a zero-source contract sensitivity. A diesel heater, renewable diesel, RNG or hybrid does not satisfy the operational endpoint. Hydrogen remains outside the electric count until a zero-emission source is certified. Existing trolleys satisfy the drivetrain condition, subject to feeder feasibility. TransLink's 2040 target makes 2033 an accelerated counterfactual.[10, 18]

The decision unit is year y, class k, depot d, trip i, vehicle v and charger interval t. Classes include standard, articulated, highway coach, double-deck, community shuttle and trolley. The service target is the recovered fall timetable; January–August 2023 departures are not available. We do not reduce that recovered service to improve feasibility.

2 Evidence and baseline audit

The 2025 budget's 2023 actual service table includes SeaBus in its published transit subtotal. We subtract 13,713 SeaBus hours, 185,370 km and 71,367,323 capacity km to obtain the fixed-route bus values in Table 1; the source's five operating-group rows reconcile exactly in the accompanying audit. The implied service-km-weighted average capacity is $5{,}715{,}791{,}401/106{,}868{,}324 = 53.48$ passenger spaces per bus-km, an aggregate calibration target rather than a standard-bus specification. The earlier statutory report rounds bus hours to 5,577,000, yielding a slightly different 98.013% route coverage check.[14, 30]

The fall GTFS is preserved at commit `eb584b1` of an independent public mirror, and every local source file matches its immutable Git blob. Its first-stop distance field is absent for every trip, so exact trip length is unknown; a monotonic shape-distance envelope gives a median interval width of 0.432 km, about 4.3%, and a maximum 26.309 km. We use the upper bound only for conservative energy screening and never fill the exact field. The annual CSV has 191 route identifiers while the feed has 232 route short names with trips because the annual table groups some branches and omits discontinued service. Both are retained with that version difference. The GTFS does not supply vehicle ids, ages, depot links, battery state or an as-operated record. Route and arrival data are used by permission of TransLink, which assumes no responsibility for their accuracy or currency.[2]

The 2018 official service guidelines define peak load factor as average peak passengers divided by practical vehicle capacity. Their reference capacities are 24 (mini-bus), 47 (standard trolley), 50 (standard bus or coach), 70 (articulated trolley), and 75 (articulated bus). Dividing each positive 2023 time-bin load by its load factor reproduces one of these values in 8,705 of 9,371 cells; a further 344 cells give exactly 85, consistent with TransLink's separately described 80–86-seat double-deck vehicles. Thus 9,049 cells (96.56%) match six capacity clusters, and 322 do not. Across 180 routes with positive cells, 34 show more than one cluster or an unmatched cell. This is a useful *aggregate vehicle-class diagnostic*, not a trip-level vehicle assignment: capacities are historical averages, the 85-person class is inferred, and mixed or differently averaged cells cannot establish an individual bus's capacity. The reproducible cell-level calculation and every unmatched value are retained in

2023 observation	Value	Source/status
Published route identifiers	191	official CSV[20]
Fall GTFS bus trips; service patterns	54,497; 25	`23SEP_20231020`[2]
Maximum trip day; source blocks	22,494; 1,756	13 October 2023[2]
Maximum-block day	22,492 trips; 1,756 blocks	8 September 2023[2]
Representative weekday blocks; simultaneous trips	1,752; 1,087	14 November 2023[2]
Route service hours	5,466,190.5	official CSV[20]
Actual fixed-route bus hours, excluding SeaBus	5,575,772	2023 actual[14]
Actual fixed-route bus service km	106,868,324	same source[14]
Actual fixed-route bus capacity km	5,715,791,401	same source[14]
Route-file share of exact bus hours	98.035%	calculated
Route/time-bin frequency rows; nonmissing	6,261; 6,041	official CSV[31]
Route/direction/time-bin load rows; nonmissing	11,659; 9,371	official CSV[32]
Revenue-fleet operating GHG, all modes	112,591 t CO_2e	2023 report[33]
CNG fleet count	299	2023 report[33]
Trolley plus battery-electric buses	280	Dec 2023 notice[8]
Battery-electric buses	about 5	Dec 2023[9]
Trolley buses	262	Feb 2023 notice[7]
2022 detailed CMBC fleet snapshot	1,711	historical anchor[34]
2023 stock scenario; sensitivity	1,725; 1,711–1,733	not a measured bound[34]
2025 active CMBC fleet	1,733; 22 battery, 261 trolley, 1,450 other	18 Nov 2025[34]

Table 1: The recovered fall timetable supplies source-declared trips and blocks. Trip and block extrema are selected separately. The 2023 stock value is a scenario anchor because the dated snapshots do not establish an end-2023 register; the annual route table and fall feed also use different scopes.

the data supplement.[21, 22, 32]

Capacity must also be checked on the replacement vehicle. Current TransLink operating limits are 20 passengers for community shuttles, 77 for 40-ft buses and 120 for articulated buses; the 2018 planning values are lower and distinguish seating from practical capacity.[27, 21] Manufacturer examples span 68 passengers for a Nova LFSe+, 88 for a current 40-ft Xcelsior CHARGE NG, 118 for the 2023 XE60 specification and 123 for the current XE60. A smaller electric bus therefore requires another departure vehicle or a different model when the baseline departure has 77 or 120 spaces.[28, 29]

The 2023 accountability report also records 112,591 tonnes CO_2e of *revenue-fleet operating* emissions, comprising 110,402 Scope-1 and 2,189 Scope-2 tonnes; it separately reports 33,718 tonnes of biogenic CO_2 from renewable fuels. The revenue-fleet number covers bus and non-bus modes, excludes vehicle and station manufacture, and reflects that year's renewable-fuel mix (96% of purchased bus natural gas was renewable). It is therefore an independent accounting check and an upper bound on the included bus operating inventory, not a substitute for the 2024–2033 same-service, life-cycle normal-renewal baseline.[33]

A later 18 November 2025 CMBC allocation table provides a retrospective fleet-profile check: 1,733 active buses at seven sites, including 261 trolleys, 22 battery buses, 535 hybrids, 365 CNG, 379 diesels and 171 gasoline shuttles; 63 more are in storage. Its manufacture-year counts reconcile to 1,733 and imply 392 buses aged 0–4 years, 543 aged 5–9, 91 aged 10–14, 705 aged 15–19 and two older than 20. It is not substituted for the 2023 register. Combined with the observed 2023 trolley and zero-emission counts, we select 1,725 for a stock scenario and 1,711–1,733 for sensitivity; these are not observed stock or proven bounds. About 285 vehicles were built in 2023–2025; a strict 2033 endpoint would retire many of them at age 8–10, so the cost model must include early replacement.[34, 8]

The 2020 TransLink fleet transition plan records 1,672 CMBC vehicles, including 188 standard and 74 articulated trolleybuses.[10] These are historical values, not a verified 2023 fleet register. The February 2023 notice gives 262 trolleys, while a December official article says approximately 275 trolleys and five BEBs. A same-day release gives only their combined count, 280. Subtracting February's 262 from December's 280 would imply 18 BEBs, but this crosses reporting dates and conflicts with the explicit December article. The retained count of 18 is only a scenario parameter, not evidence of deliveries.[7, 8, 9] A June 2023 Board report lists 2023–24 procurement projects for 46 articulated hybrids, 15 standard BEBs, 50 CNG buses, 168 community shuttles and 25 double-deck buses; it records the 25 double-deck buses as accepted on a Q2 2023 timeline, while the other group totals were mostly outstanding. These are procurement states, not a year-end fleet roster.[12] A valid 2033 purchasing table still requires an age, class and depot register.

A separate June 2023 TransLink presentation places four BEBs and 262 trolleys in service *at that date*. It gives screening prices of approximately CAD 1.2 million per BEB and CAD 0.46–0.61 million per on-route charger. Its then-current capital concepts put Hamilton Route 100 charging at CAD 2.3 million, Port Coquitlam plus ten on-route chargers at CAD 46 million, and a Marpole depot sized for 340 BEBs at CAD 500 million. A newly built 500–600-bus-equivalent depot was screened at CAD 1.2–1.8 billion including land and development. These are 2023 planning envelopes with different scopes, *not* delivered item prices or additive line items; their commissioning dates changed in later plans. They supply bounded infrastructure cases while site-specific electrical capacity and construction cash flow remain missing.[13, 18]

A 2025 capital-program audit narrows the age uncertainty: 188 standard trolleys entered service in

2006–07 and 73 articulated trolleys plus one pilot were expected to retire in 2028–29. It identifies projects tied to 57 diesel-hybrids and 50 CNG buses reaching end of life in 2023, 126 diesel-hybrids in 2024, 36 diesel-hybrids and nine highway coaches in 2025, and 39 articulated diesels in 2026. The plan specifies 57 BEBs for 57 retiring hybrids, seven BEBs for six retiring hybrids, and 45 articulated BEBs for 39 retiring diesels in three separate programs. These are retrospective planned cohorts, not confirmed 2023 stock, deliveries, or a route-feasible 2033 solution. The 57-BEB project's CAD 106.846 million gross capital budget includes CAD 102.498 million of reported external funding; divided by 57, its *gross project budget* is CAD 1.874 million per planned bus. The other identified BEB projects yield CAD 1.867 million per planned standard bus and CAD 3.037 million per planned articulated bus on the same budget/count arithmetic. These are project envelopes, not delivered bus-only unit quotes. The accompanying cohort audit reproduces the ratios and divisions from the published table.[14]

As a bounded funding illustration using only the 57-BEB project's gross budget, assuming *all* CAD 106.846 million is grant-eligible, 0%, 25% and 50% hypothetical grants leave agency payments of CAD 106.846, 80.135 and 53.423 million, respectively. The gross project budget remains CAD 106.846 million in each case. These are undiscounted funding splits of one later-reported program, not incremental economic scores: eligibility, spending dates, and the same-service conventional renewal budget are unresolved. The actual CAD 102.498 million external-funding entry is a distinct retrospective budget fact and does not override the problem's 50% scenario limit.[14]

A subsequent 2026 plan changes the retrospective picture further. A named project now proposes 39 60-foot diesel buses, 64 in-route-charged 40-foot BEBs and 153 depot-charged 40-foot BEBs for 236 retiring diesels: 256 planned replacement vehicles, or 1.085 per retirement. Another cohort proposes 31 BEBs for 27 retiring hybrids, a 1.148 ratio. The 2025 plan's 45 articulated BEBs for 39 retiring diesels is an earlier description with possible cohort overlap; its count is *not* added to the later project. The 2025 plan separately names 50+84+30=164 CNG replacements for 2023–2025 retirement cohorts. Thus these distinct named programs plan $164 + 39 = 203$ non-electric replacement vehicles associated with 2023–2026 retirement cohorts. If procured and still operating, all 203 would need accelerated removal to achieve the strict 2033 endpoint. This is a planning-exposure count, not a verified delivery count or a complete 2033 fleet requirement.[14, 16]

3 Fleet and service model

3.1 Decision variables and normal-renewal baseline

Let $x_{ykd} \in \mathbb{Z}_{\geq 0}$ be new buses delivered to class k and depot d; r^p_{ykda} vehicles retired from age cohort a; h_{ydq} charger positions; and $z_{iv} \in \{0, 1\}$ a trip assignment. Let N^p_{ykda} be stock and $p \in \{\text{conventional}, \text{battery}, \text{trolley}\}$ with disjoint propulsion categories. A purchase enters only at age zero. Historical expenditure is sunk; early retirement enters through losing residual value or obtaining a sale receipt.

For each class and depot, without used-vehicle purchases or inter-depot transfers,

$$N^p_{y,k,d,0} = x_{y,k,d}, \qquad N^p_{y,k,d,a} = N^p_{y-1,k,d,a-1} - r^p_{y,k,d,a} \quad (a \geq 1), \tag{1}$$

with $0 \leq r^p_{y,k,d,a} \leq N^p_{y-1,k,d,a-1}$ and integer stocks. Transfers or used purchases would add explicit inflow and outflow terms. No diesel, hybrid, CNG or hydrogen propulsion or non-electric accessory

may operate on a fixed-route trip in 2033. Purchases are not one-for-one replacements: schedule, range, spare and charging evidence would determine the physical count.

3.2 Hard service and technical constraints

For every recovered fall service pattern p, every trip $i \in I_p$ is retained at its route, direction and departure time. Multiple smaller buses may serve one departure when each is compatible and their passenger allocations do not interact:

$$\sum_{v \in V_i} z_{iv} \geq 1 \qquad (i \in I), \tag{2}$$

$$\sum_{v \in V_i} Q_v z_{iv} \geq Q_i^0 \qquad (i \in I). \tag{3}$$

Here Q_i^0 is a capacity proxy, not observed passenger demand. The route–direction peak is used only as an aggregate diagnostic because a time-bin maximum need not coincide with a particular trip set. Accessible trips require accessible vehicles; spare stock and class compatibility remain constraints.

The source block is a duty hypothesis. For consecutive trips $i \prec j$ on one vehicle, $s_j - e_i \geq \tau_{ij}^{road} + t_{ij}^{turn}$. Using straight-line distance and a generous 50 km/h with a five-minute turn, 1,249 of 20,679 weekday transitions cannot be completed even before road detours; zero gap therefore does not by itself invalidate a block, and a block ID does not certify it either. On the terminal-linked Route 100 sample, all 200 transitions pass this necessary screen: 223 trips and 23 source blocks become 25 duty fragments under 300 km, with an independent simultaneous-trip lower bound of 14. This sample preserves those trips but still lacks garage, charger and road-time certificates.

The maximum trip day has 22,494 trips; 8 September has the maximum block count and 22,492 trips. The annual output also reproduces 5,575,772 hours, 106,868,324 service km and 5,715,791,401 capacity km within documented measurement differences. These calibrations cannot substitute for departure, block and site constraints.[2, 14]

The executed range screen uses

$$d_b^L = \sum_{i \in b} \ell_i^L, \qquad d_b^U = \sum_{i \in b} \ell_i^U, \qquad n_{bk}^{L,(j)} = \left\lceil \frac{d_b^L}{R_k^{(j)}} \right\rceil, \qquad n_{bk}^{U,(j)} = \left\lceil \frac{d_b^U}{R_k^{(j)}} \right\rceil. \tag{4}$$

For the representative weekday and $R = 300$ km, the two sums are 1,698 and 1,741. Across all 25 recovered patterns, upper-distance requests range from 762 to 1,745. These are fixed-range, no-mid-duty-recharge distance screens. Deadhead, terminal charging and task reuse can move the physical requirement in either direction, so neither endpoint is a vehicle lower bound.

Each vehicle's assigned trips and charging intervals cannot overlap. Location-to-location deadhead times must fit between events, including departure from and return to its depot. Driving to or from service also removes a vehicle from the available depot spare count. For battery vehicles, energy follows

$$e_{v,t+1} = e_{vt} + \eta_v c_{vt} - \epsilon_{vk}^{\text{traction,winter}}(\ell_{it} + \ell_{vt}^{\text{deadhead}}) - E_{vt}^{\text{accessories}}, \tag{5}$$

$$\underline{s}_v B_{v,y} \leq e_{vt} \leq \bar{s}_v B_{v,y}, \tag{6}$$

$$c_{vt} = \textstyle\sum_d P_{vdt} \Delta t, \qquad e_{v,d,T} = e_{v,d+1,0}, \qquad e_{v,N,T} \geq e_{v,1,0}. \tag{7}$$

Here c_{vt} is grid-side energy and $B_{v,y}$ includes degradation. The daily validator accepts one day; the periodic wrapper carries ending SOC into the next day. An unmeasured winter accessory load

makes a schedule `UNVERIFIED`. Charger occupancy, power and opening year impose

$$\begin{aligned} 0 \le P_{vdt} \le \bar{P}_{vd} u_{vdt}, \qquad & \sum_{d} u_{vdt} \le 1, \\ \sum_{v} u_{vdt} \le H_{dt}, \qquad & \sum_{v} P_{vdt} \le P^{\text{grid}}_{dt}, \qquad H_{dt} = 0 \text{ until commissioned.} \end{aligned} \tag{8}$$

Conventional baseline vehicles instead require a fuel budget, with jointly calibrated distance and elapsed-duty rates a_v, b_v in a consistent fuel unit (L or kg):

$$F_{v,t+1} = F_{vt} - a_v \ell_{vt} - b_v \Delta t, \qquad F^{\text{reserve}}_{v} \le F_{vt} \le F^{\text{tank}}_{v}. \tag{9}$$

The implemented check assumes no refuelling within a duty; the time rate includes stationary consumption and must not duplicate the distance rate. Missing fuel data leave the normal-renewal comparator unverified. Such vehicles may serve intermediate years but cannot satisfy the strict 2033 endpoint.

Trolleybuses use a separate wired-network check. Each trip, deadhead and powered layover needs a certified connected path and an operating feeder. Concurrent demand must obey $\sum_{v \in V_{ft}} P_v \le P^{\text{firm}}_{f}$, where firm power is available after other loads and reserve. The code conservatively reserves each vehicle's measured all-in peak draw on every feeder traversed for the entire moving segment; stationary accessories use their separate demand. A failure of this sufficient power certificate calls for finer feeder timing before declaring physical infeasibility. Missing path or feeder records leave existing trolley operation unverified, without assigning these vehicles fictitious batteries or buying their capacity again.

Depot bus parking capacity, charger footprint, charging efficiency, construction lead time and route-specific winter testing must also pass. A charger site may be the home depot or a route terminal, provided the location and travel-time checks pass. The accompanying validator checks the supplied trip, energy and site records, returning `UNVERIFIED` for missing essential inputs; it cannot certify a city-wide solution from time-bin averages.

3.3 Explicit assumptions and validation

Five assumptions define the evidence boundary. (A1) Every recovered fall departure and direction is a service obligation; the representative weekday is a planning case. (A2) The values 24, 47, 50, 70, 75 and 85 are capacity proxies. (A3) Reliable range is a technology scenario. (A4) The 1,725-bus stock and 18 existing BEBs are unverified scenario parameters, with the stock and retained-BEB grids varied jointly. (A5) Price envelopes are rebased to 2023 CAD using Bank of Canada CPI; CPI is not a construction index, so the 3% rate is a real-rate scenario. Validation includes a 100-kWh bus whose 60-km day passes once but fails on the second day without charging, then passes when return charging restores the cycle. The local pipeline does not certify a city-wide schedule.

3.4 Optimization and executed decomposition

The complete master problem would link purchases and age cohorts to available vehicles, trip assignments, daily SOC, site commissioning and the renewal comparator:

$$\min C^{\text{gross}} \quad \text{s.t.} \begin{array}{l} \text{service, block and capacity;} \\ \text{SOC, charger and feeder;} \\ \text{spare, inventory, commissioning and endpoint.} \end{array} \tag{10}$$

This is a design statement, not a solved city MILP. The station, depot, transfer and age data are incomplete, so no Pareto frontier is claimed. We execute four bounded pieces: fall block and range screens; a Route 100 terminal duty sample; proportional inventory cost scenarios; and age-constrained Chicago/London cohorts. The validator returns `UNVERIFIED` when a required event or site record is absent.[35, 36, 37]

4 Costs, funding and environmental accounting

Let K_y include buses, initial batteries, chargers, depot civil work, grid upgrades, design and training. Let O_y include energy, maintenance, replacement batteries, drivers, demand charges and site upkeep. Let S_y be realized sale receipts and R_{2033} all vehicles and durable infrastructure remaining at the horizon. Public envelopes are rebased with the Bank of Canada total CPI to 2023 CAD; the 3% rate is then a real-rate scenario.[6] CPI is economy-wide, so construction costs remain a material source of uncertainty.

$$C^{\text{gross}} = \sum_{y=2024}^{2033} \frac{K_y + O_y - S_y}{(1+r)^{y-2023}} - \frac{R_{2033}}{(1+r)^{10}}, \tag{11}$$

$$\Delta C = C^{\text{gross}}_{\text{electric}} - C^{\text{gross}}_{\text{renewal}}. \tag{12}$$

For grant rate $f \in \{0, 0.25, 0.50\}$ and eligible expenditures E_y, $C^{\text{agency}} = C^{\text{gross}} - \sum_y f E_y/(1+r)^{y-2023}$. A grant changes who pays, not gross buses, copper, electricity or labour. Local employment and procurement impacts require an additional regional economic model and are reported separately to avoid double counting.[38]

The regional layer records direct local procurement L_y, wages W_y and job-years J_y. Any induced value added uses a regional input–output multiplier m_y on labour and supplier value added, $V_y^{\text{local}} = m_y(L_y + W_y)$, not on the entire gross construction outlay. Feasible charging capacity, reliability and terminal asset value enter the fleet model through served trips, avoided extra vehicles, bill savings and residual value. Construction spending is not counted again as a benefit.

Define a life-cycle greenhouse-gas inventory over equal service:

$$G = G_{\text{vehicle+battery}} + G_{\text{infrastructure}} \tag{13}$$

$$+ \sum_y (G_{\text{fuel,y}} + G_{\text{electricity,y}} + G_{\text{maintenance,y}}) - G_{\text{end of life}}. \tag{14}$$

The executed inventory includes vehicle and battery production, operating energy and maintenance. It excludes depot construction and end-of-life credits. British Columbia's 2023 integrated-grid factor, 0.0113 kg CO_2e/kWh, is used as an annual average; a zero-source contract sensitivity

sets electric operating emissions to zero without claiming that a certificate exists.[39] NO_x, particles, electricity and battery production remain separate. The denominator covers scored conversion service only because common trolley and retained-BEB terms are not measured on both sides.

The requested scores are

$$S_C = 100\frac{C_{\text{renewal}} - C_{\text{electric}}}{C_{\text{renewal}}}, \qquad S_G = 100\frac{G_{\text{renewal}} - G_{\text{electric}}}{G_{\text{renewal}}}. \tag{15}$$

Negative values remain negative. The economic score is based on *gross* cost, with subsidized agency cost shown beside it; otherwise the 50% grant would artificially change the societal merit ranking.

5 Observed and conditional results

5.1 Fall 2023 block and range screen

The recovered feed contains 54,497 bus trips, 25 service patterns and 1,684,516 bus stop-time records after excluding rail, SeaBus and HandyDART. On 14 November 2023 it schedules 22,431 trips and 1,752 source-declared blocks. At the busiest instant, 1,087 trips are simultaneous. The maximum trip day is 13 October with 22,494 trips; 8 September also has 1,756 blocks and 22,492 trips. These are planned duties, not an as-operated registry.[2]

For energy screening, we classify blocks by 24/47/50/70/75/85 passenger-space proxies. The 47/70 signatures remain only a trolley-route proxy; the GTFS has no propulsion field. The 57 weekday double-deck blocks are excluded from overnight arithmetic. At 300 km, the representative weekday yields 1,698–1,741 requests for 1,502 participating blocks. At the 149/150-km electric-heat band it yields 2,444 requests, 62.7% more. Across all 25 recovered patterns, upper-distance requests range from 762 to 1,745. Route 100 provides the smallest auditable duty sample: 223 trips, 23 blocks and 25 terminal fragments, bounded below by 14 simultaneous trips. It passes the direct-transfer screen but lacks depot and charger records.

Proxy class	Baseline blocks	Additional sequential requests, lower–upper distance 149/150	184/185	250/251	300/300	350/350 km
40-ft battery proxy	1244	727–788	525–577	295–333	167–202	73–96
60-ft battery proxy	258	135–154	110–119	54–66	29–37	18–25

Table 2: 2023-11-14 service with 1,752 published duty blocks and 1,087 simultaneous trips. Each cell is the lower–upper count from the trip-distance interval under a fixed no-mid-duty-recharge range. Capacity signatures are route-level proxies, not observed bus assignments. The 40-ft battery proxy contains minimum/standard capacity classes and unclassified blocks; 47/70-seat signature blocks are retained only as a trolley-route proxy, and 85-person double-deck blocks are unassessed. The interval is neither a physical vehicle bound nor a procurement quantity.

Only 3 of the 20,679 weekday block transitions provide at least 60 minutes of gap; among the 4,850 route-changing transfers, only 2 do, and none provides 90 minutes. A generous 50 km/h plus five-minute screen rejects 1,249 transitions even before road detours. Blocks therefore remain duty hypotheses requiring terminal charging, exchange or documented compatible routing.

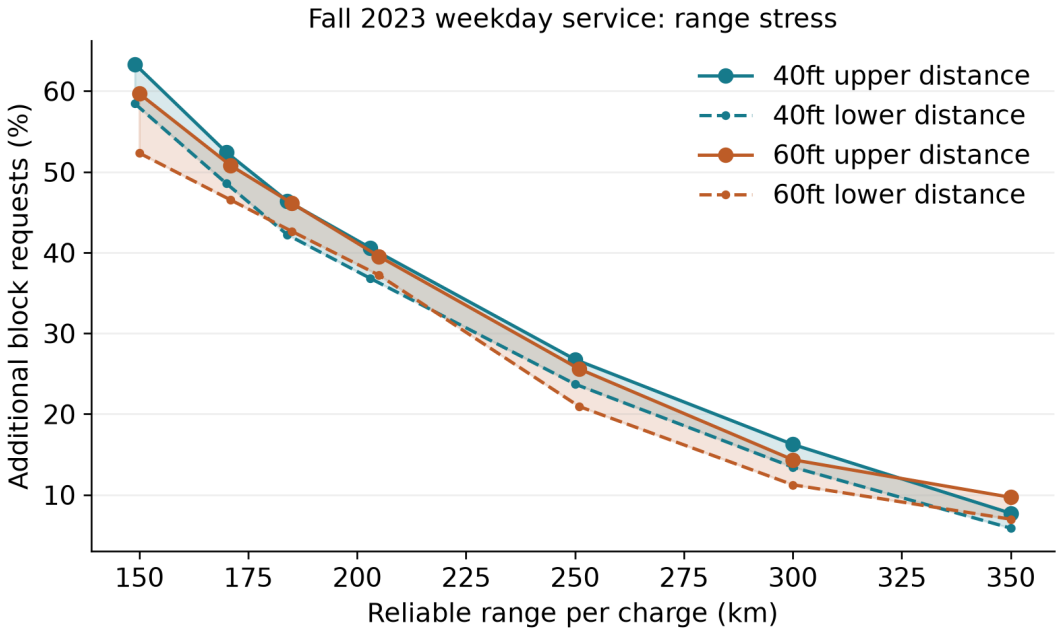


Figure 1: Lower and upper trip-distance ceilings under fixed reliable range. The interval is a sequential-request screen; it does not include road transfer time, garage return, charger occupancy or spare buses. Capacity signatures are not certified vehicle assignments.

5.2 Conditional ten-year cost and emissions

We evaluate an accelerated stock scenario rather than relabel a full replacement as ordinary renewal. A uniform 17-year fleet renews 10/17 of its positions in a decade; 1,388 non-deck fossil buses yield 816 baseline purchases. The upper-distance request ratio is projected onto an assumed 1,406-bus stock, selecting 1,630 positions and 1,612 route purchases after 18 assumed BEBs. This is neither a lower nor upper vehicle bound. Prices are constant 2023 CAD after CPI rebasing: CAD 1.2 million hardware screen, CAD 1.874 million standard and CAD 3.037 million articulated project envelopes, and CAD 0.923 million for the baseline CNG project.[13, 14, 15]

Marpole's CAD 500 million/340-bus ratio prices the base infrastructure; the CAD 6.52-billion multi-site envelope that includes service expansion is the high sensitivity.[13, 18] Route-vehicle delivery is scheduled for 2029–33, aligned one year before staged sites first become available. Construction spending starts in 2024, but no electric-service share is credited until 2030. Vehicle residual uses a 17-year life. Driver hours, common trolley renewal and the 57 double-deck buses remain outside this generic sub-boundary.

The generic cost case omits initial retained-asset terminal values, early-retirement receipts and infrastructure terminal values. Their signs do not justify calling the result a conservative bound, so they are evaluated separately in the class asset table. The high-vehicle case includes the broad multi-site envelope and is not a fixed-service upper bound.

The generic non-deck constant-2023 present value is CAD 4.30 billion against CAD 2.27 billion for steady-state renewal (-89.2%). The high case reaches CAD 7.98 billion (-251.0%). Symmetric 25/50% grants give agency scores of $-49.5/-9.8\%$; grants change affordability, not gross resource cost. These figures are conditional: range is a 2020 sensitivity, depot power is absent, and no 2023 age/depot register exists.

Scenario	Gross PV (CAD bn)	Score (%)
Conventional renewal	2.275	0
Electric, base infrastructure	4.303	-89.2
Electric, screening high	7.984	-251.0
Agency PV and score, grants 0/25/50% (base plan)		
	4.303 / 3.401 / 2.499	-89.2 / -49.5 / -9.8

Table 3: Conditional generic non-deck inventory scenario in constant 2023 CAD. The 1,698–1,741 distance requests are proportionally projected to 1,630 positions and 1,612 route-purchase parameters after 18 assumed existing BEBs; this is not a vehicle or dispatch certificate. The 57 double-deck buses and 262 trolleys are outside this sub-boundary. The baseline renews 10/17 of unknown-age fossil positions.

Baseline operating mix	g CO_2e/km	Renewal (kt)	Electric (kt)	Score (%)
renewable fuel mix	302	376.2	442.4	-17.6
rng heavy	960	960.4	874.8	8.9
weighted mix	1233	1202.8	1054.1	12.4
fossil diesel	1580	1510.9	1282.1	15.1

Table 4: Conditional generic non-deck ten-year emissions over a partial production/operation/maintenance boundary. No conversion is credited before 2030; readiness shares reach 25/50/85/100% in 2030–2033, and conventional kilometres remain in every prior year. Depot construction and end-of-life credits are excluded. The renewable mix assumes 80% renewable-diesel reduction plus 96% RNG for CNG; RNG-heavy uses RNG but fossil diesel; weighted mix uses the 2020 propulsion mix.

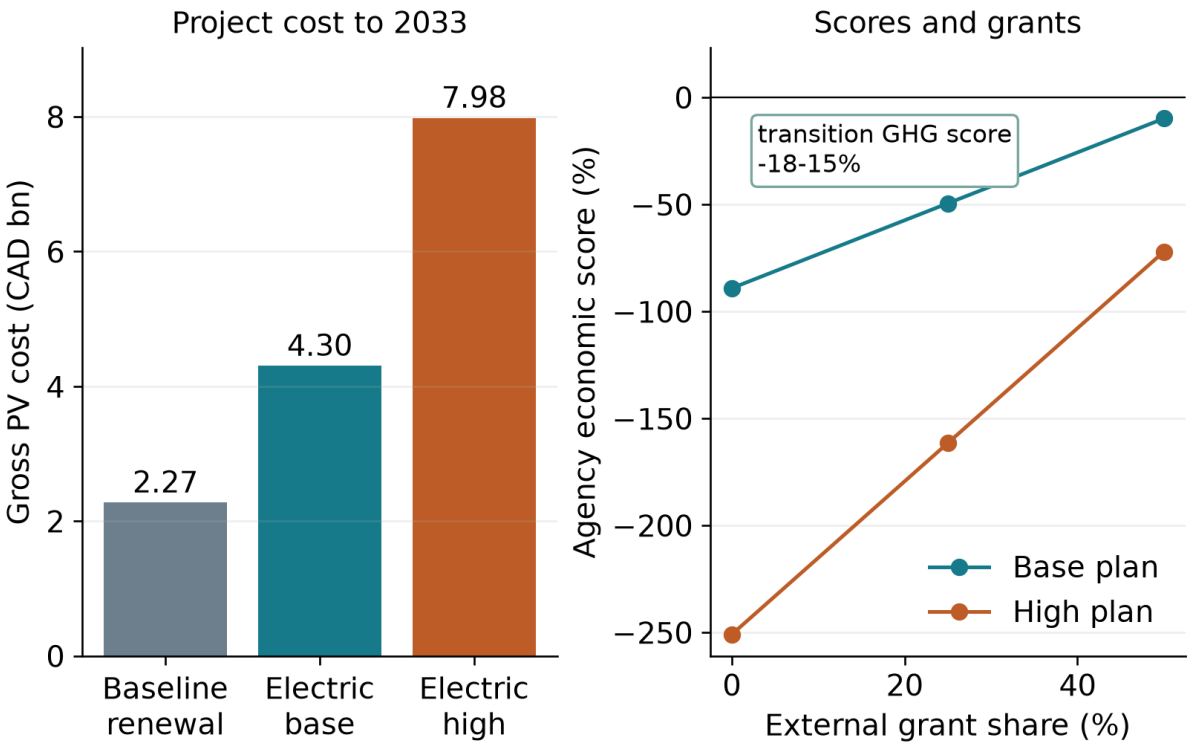


Figure 2: Conditional generic non-deck economics and funding; curves show agency scores. Constant 2023 CAD and facility readiness are applied to both the money and service ledgers. The boxed range is the generic partial production/operation/maintenance result, which includes a negative renewable-fuel case; depot construction and end-of-life credits are excluded.

A class inventory uses 1,419 heavy positions, 57 double-decks and 189 community shuttles. It includes 1,401 heavy, 57 deck and 189 shuttle battery purchases, plus 262 common trolley renewals. Common trolley assets are in both financial paths. Central constant-price PV is CAD 4.43 billion versus CAD 2.62 billion (−68.8%); symmetric 0/25/50% grants give −68.8/ − 38.8/ − 5.5%. The high vehicle case is CAD 4.73 billion (−80.2%). A 700-kWh double-decker leaves 17 of 57 weekday duties short of one overnight charge; 827 kWh leaves eight. A 150-kWh shuttle leaves 49 of 101 short. Terminal or en-route charging is therefore a hard gate, while the 1,927 electric-asset endpoint is only stock arithmetic.[12, 23, 24, 25, 26, 18, 14, 16]

Vehicle cost case	Renewal PV	Electric PV	Score (%)
Low	2.627	4.412	-68.0
Central	2.625	4.430	-68.8
High	2.623	4.726	-80.2
Agency score, common trolley grant symmetric in both paths			
Grant 0/25/50%			-68.8/-38.8/-5.5

Table 5: Class-specific inventory scenario in constant 2023 CAD (billion; else percent). The electric and renewal paths both include common trolley renewal. The environmental percentage is a scored-conversion subset, while cost includes common trolley capital; the two percentage denominators are therefore different.

Sale recovery	Infra life	Renewal PV	Electric PV	Score (%)
0%	excluded	2.143	4.423	-106.4
0%	30	2.143	3.140	-46.5
10%	excluded	2.143	4.406	-105.6
10%	30	2.143	3.122	-45.7
25%	excluded	2.143	4.380	-104.3
25%	30	2.143	3.096	-44.4

Table 6: Asset-boundary cases in billion 2023 CAD. Retained 2023 assets now use the same straight-line life and uniform age assumption in both paths; early conventional retirement receives only the stated share of remaining replacement-cost value. Published infrastructure costs are also shown with a 30-year terminal-value case. These are scenario values, not accounting book values or observed resale bids.

The purchase-only residual is incomplete because the baseline retains initial assets. A symmetric asset memo applies the same 17-year straight-line proxy to retained stock, newly purchased vehicles and early-retirement receipts. With no infrastructure terminal value, the score is −104 to −106%; with a 30-year infrastructure value it is −44 to −46%; across 20–40 years and 0–25% resale recovery, −41 to −53%. The range remains far from zero. Uniform initial ages and replacement-cost proxies are scenarios, not book values.

A per-class annual ledger drives O&M, grid energy and emissions. Site readiness is zero in 2024–2029, then 25, 50, 85 and 100% in 2030–2033; conventional kilometres remain in every prior year. Use emissions are $D \sum_y [(1 - \alpha_y)(g + 0.0524) + \alpha_y(0.019 + 0.0675)]/1000$ tonnes. The generic case scores −17.6 to 15.1%: renewable fuel is negative because manufacturing dominates when

conversion begins late; the other fuel cases are 8.9, 12.4 and 15.1%. The class case scores −14.3 to 16.2% and avoids −55.5 to 257.6 kt against 387–1592 kt baseline emissions. Unknown shuttle bodies are omitted on both sides; standard-bus glider proxies remain for heavy and deck vehicles. Both percentages cover scored conversion service, excluding common trolley and retained-BEB activity, so they are not whole-fleet life-cycle scores. Depot construction and end-of-life remain excluded. The annual tailpipe proxy is 126–408 t NO_x, 1.11–8.34 t $PM_{2.5}$ and 2.34–14.18 t PM_{10}; its fuel-by-km split is uncertain.[19, 18, 43, 44, 45]

At 1.37–2.0 onboard kWh/km and 90% charging efficiency, converted non-deck service needs approximately 135–197 GWh/year from the grid; the 1.6 kWh/km central value gives 158 GWh/year. The 1,612 new 335-kWh batteries contain about 31.3 kilotonnes of modeled LFP-production CO_2e using the transferred factor. These are scale checks, not a site load profile or a Vancouver battery specification.

As a historical benchmark, the route table still covers 98.035% of independently reported actual bus hours. The 2019-CAD and 2024-O&M reference cases retain their original narrower boundaries. These checks and the fixed-service policy case support the fleet decision; only an operator trip, garage, charger and age register can certify every 2033 duty.

For a transparent historical parameter check, the 2020 plan lists a 2019-CAD diesel bus at CAD 630,000, depot BEB at CAD 1,110,000, 50-kW charger position at CAD 298,000, 0.60 L/km diesel, 1.4 kWh/km electricity and 0.02 L/km heater fuel.[10] At 68,000 km/year the historical BEB uses 95,200 kWh plus 1,360 L of fuel, so its heater fails the strict 2033 endpoint. The CAD 298,000 charger quotation already includes design, permitting, project management, financing and contingency and must not be repriced. This is not a 2023 bid; TransLink's 1.15 depot replacement ratio also shows why one-for-one purchase is unsafe.

For a separate electric-only 40-foot reference, 2019-CAD lifetime costs are CAD 4.86/km for in-route BEB versus 4.62 diesel, 4.77 hybrid and 4.15 CNG; the three economic scores are −5.19%, −1.89% and −17.11%. These are single-vehicle comparisons, not fleet scores. Using 1.6 kWh/km, 0.60 L/km diesel, 2.630 kg CO_2e/L and the BC grid factor gives a 98.85% operational-only reduction, or 106.07 t CO_2e/year at 68,000 km. Vehicle identities, as-operated execution and site capacity remain unavailable, so this excludes manufacturing and fuel supply.[10, 11, 40, 39]

The 2024 ZEFTP offers a separate retrospective check. Its operating-and-maintenance costs are 3.30, 1.97, 2.77 and 2.28 CAD/vehicle-km for diesel, hybrid, CNG and BEB, respectively. On that *O&M-only* boundary, BEB scores are +30.91%, −15.74% and +17.69% against the three alternatives; vehicle, battery and charger capital are absent. Its electric-heater BEB is reported at 19 g CO_2e/km, versus 1580 for fossil diesel, 1420 for fossil hybrid, 1350 for fossil CNG and 250 for renewable CNG. Fuel mix therefore changes the operational environmental comparator. These 2024 values are neither 2023-known procurement inputs nor full life-cycle 2024–2033 totals; they cannot be added to the older 2019-CAD costs.[18]

An independent 2024 ICCT e-bus life-cycle method supplies transferable *sensitivity factors*: 6.6 kg CO_2e per kg of vehicle glider/powertrain, 58 kg CO_2e per kWh of LFP battery, and 52.4/67.5 g CO_2e/km for diesel/BEB maintenance. Its 12–15 m example uses a 335-kWh battery and glider masses from Latin American buses, so these are not observed TransLink specifications. Its own system boundary excludes charging infrastructure and bus recycling. We retain separate terms for these missing components and do not substitute this partial method for a completed Vancouver life-cycle inventory.[19]

Facility timing is also consequential. A December 2020 program listed 677 planned BEB purchases

and CAD 246.4 million of depot/charging investments, but subsequent 2024 review targeted Port Coquitlam for over 100 BEBs in 2027 and Marpole for roughly 340 in 2028. It estimated six to eight years to deliver charging at existing transit centres and recommended commissioning six to twelve months before bus delivery. These are planning and later-review quantities, not 2023 installed charger counts. The 2024 screening capital envelope includes service expansion and has a stated −50% to +100% accuracy range, so assigning it wholesale to our fixed-2023-service scenario would misstate incremental cost.[11, 18]

The 2020 program arithmetic checks 677 planned BEB awards: CAD 236.9 million vehicle premium plus 246.4 infrastructure minus 96.3 operating savings gives CAD 387.0 million undiscounted. A later plan instead lists 84 CNG buses for a 2024 hybrid-retirement cohort, showing that the pathway was revised. These source-period amounts cannot be converted into 2024–2033 cost, grant or GHG scores without aligned dates and fleet stock.[11, 14]

The historical depot analysis supplies a direct *winter feasibility stress test*. With 660-kWh articulated batteries and fuel heaters, it projected depot-dependent replacement ratios of 1.13–1.55 and warned that electric-only heat could require roughly 10% more buses. Applied mechanically to the later capital project's 39 retiring articulated diesels, $\lceil 39(1.10)r_d \rceil$ gives 49–67 BEBs across hypothetical depot allocations, compared with its 45 planned depot-charged BEBs. The 4–22 difference is a *historical-technology sensitivity*, not a measured shortage: the 39 actual depot assignments, 2026 energy specifications and block schedules are unknown. It identifies a concrete winter trial and procurement question before certifying the strict 2033 endpoint.[10, 14]

5.3 Charging tariffs and the value of firm power

The original 2023 budget assumes diesel at CAD 1.66/L; this is a planning price, not an actual invoice. Its 32.17-million-L operational volume includes a broader system boundary and is not treated as fixed-route bus consumption.[3] BC Hydro's April-2023 tariff supplies energy, monthly demand and daily basic charges. RS1640 bills peak demand only during 06:00–22:00; RS1650's zero demand fee ends in March 2026 under that historical schedule. We therefore distinguish billed peak from physical grid power and do not extrapolate the temporary concession through 2033.[4]

For daily grid energy E, available charging hours T, site limit P, and rate-specific billed hours T_j, solve

$$\min M_j : \quad \sum_{t \in T} p_t \Delta t = E, \quad 0 \le p_t \le P, \quad p_t \le M_j \ (t \in T_j). \tag{16}$$

Power is zero outside T. If h_0 available hours are outside the billing window and $h_1 > 0$ are inside, the sharp lower bound is $M_j \ge \max\{0, (E - h_0 P)/h_1\}$. Physical feasibility also requires $E \le (h_0 + h_1)P$; a wholly nonbilled feasible window gives $M_j = 0$. The executed CPU LP matches this independent bound in every feasible case. A second stage minimizes physical peak at the optimal billed peak. Monthly bills include energy, basic and demand charges, the prior-winter eleven-period minimum-bill floor, and the 1% rider refund; taxes and capital are excluded.

We normalize to 100 *distance-equivalent units*, each at the historical 68,000 km/year reference. These are not an observed fleet or a one-for-one replacement count. Historical onboard energy of 1.37–2.0 kWh/km, assumed charging efficiencies of 85/90/95%, three common return windows and 2.5/3.5/5-MW site limits give 243 scenarios. Annual energy is spread evenly across 365 days; this relaxation does not certify winter energy, battery capacity, charger-port occupancy or actual bus availability. A meter-measured intensity would not be divided by efficiency again.[10]

Rate	Physical MW	Billed MW	Annual CAD million
1600	4.140	4.140	1.350
1640	4.140	0.000	0.905
1650 (pre-Apr 2026)	4.140	4.140	1.113

Table 7: Conditional charging bounds for 100 distance-equivalent units, 68,000 km/year each, 1.6 battery-kWh/km, 90% charging efficiency, 22:00–06:00 availability and 5-MW site power. April-2023 prices are frozen for a 365-day comparison; bills include the 1% rider refund and exclude taxes and capital.

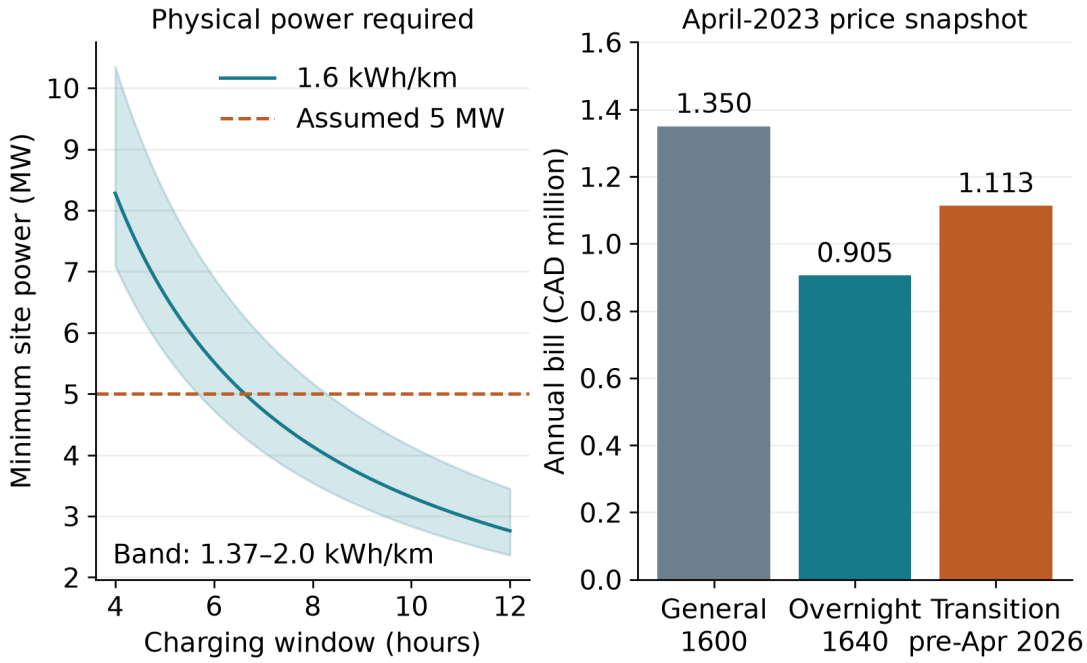


Figure 3: Physical charging-power bounds and conditional tariff bills. The left panel uses 90% charging efficiency; the right uses the central eight-hour case. A zero billed peak still requires 4.140 MW of physical power.

In the central case, an eight-hour window needs at least 4.140 MW, and RS1640 saves 32.96% of the RS1600 electricity bill. At 3.5 MW, eight-hour-only charging is impossible under the declared energy requirement. Extending availability to 20:00–06:00 restores aggregate feasibility, but RS1640 then costs CAD 1.285 million/year versus CAD 1.227 million under RS1600. The bills cross at 3.598 MW; two separate LP solutions confirm this threshold. Between 3.312 and 4.140 MW, each extra firm kW saves CAD 594/year under RS1640 in this fixed-price, fixed-window case, before paying for the upgrade. This quantifies an infrastructure contribution through feasible service and avoided bills, without crediting construction expenditure as a second benefit. Individual vehicle certification and dated future tariffs remain necessary before using these bounds in a purchasing decision.

Depot	Active	Non-trolley	50-kW scale (MW)
VTC	430	169	8.45
HTC	289	289	14.45
BTCN	107	107	5.35
BTCS	161	161	8.05
STC	266	266	13.30
PTC	235	235	11.75
RTC	245	245	12.25
Total	1733	1472	73.60

Table 8: 2025-11-18 CMBC active-fleet scale by site. This is retrospective allocation, not 2023 stock; the power column is active non-trolleys times the historical 50-kW planning allowance, not available transformer capacity.

At the 2020 planning allowance of 50 kW per depot-charged bus, the generic 1,612 positions imply about 80.6 MW and the 1,647-position class conversion about 82.4 MW of aggregate connected capacity. TransLink's 2024 long-range estimate is 78 MW for full-fleet electrification by 2040. The 2023 accelerated plan therefore has roughly the same grid-scale challenge as the published 2040 path. This is a portfolio scale check, not a site transformer rating; only a feeder and interconnection study can determine 2033 feasibility.[10, 18]

6 Ten-year roadmaps for three metropolitan areas

The transfer model separates stock arithmetic from deadline compliance. Vancouver's class inventory has facility-gated conversion shares; Chicago links its reported age bins to hard replacement deadlines; London solves age cohorts with body-type availability. Their unmatched prices, tariffs and route records prevent a single cross-city dollar or emission score.

The Vancouver inventory buys 1,647 route vehicles plus 262 common trolley renewals: 281 route vehicles in 2029, then 341–343 per year in 2030–33. No conversion service is credited before 2030. Retirements follow 25/50/85/100% readiness in 2030–33 and end at 1,927 electric assets. The 202 extra positions are a proportional scenario, not a duty-splitting consequence. A funded order and commissioned site must precede each conversion; the stock table cannot certify this timing.[18, 14, 16]

Year	Van. EV buy	Van. EV	Van. fossil	CTA buy	CTA fossil	TfL buy	TfL fossil
2024	38	280	1445	185	1665	775	6918
2025	38	280	1445	185	1480	776	6142
2026	39	280	1445	185	1295	776	5366
2027	37	280	1445	185	1110	776	4590
2028	74	280	1445	185	925	776	3814
2029	317	561	1445	185	740	776	3038
2030	343	904	1084	185	555	776	2263
2031	341	1245	723	185	370	776	1493
2032	341	1586	217	185	185	776	724
2033	341	1927	0	185	0	776	0
Total	1909	1927	0	1850	0	7759	0

Table 9: Annual stock-flow comparison. Vancouver's 1,909 electric purchases comprise an assumed 1,647 route positions plus 262 common trolley renewals. Chicago's CTA buy column is the legacy 1,850/10 linear arithmetic, retained only for comparison: its separate age test below shows a 2024 hard-deadline shortfall. TfL renews body-compatible slots under its 17-year scenario. Zero 2033 stock arithmetic is not a route, garage or charging certificate.

Chicago's year-end inventory is 1,875 buses, with 25 inferred electric slots. Keeping 185 purchases/year reaches zero conventional stock in 2033 but fails a hard age deadline immediately: using each reported bin's oldest member, 539 vehicles are due in 2024 under a 17-year cap, leaving a 354-slot shortfall. Under 14 and 12 years, the 2024 shortfalls are 871 and 901. The healthiest uniform annual capacity would be 539, 1,056 or 1,086 in those scenarios. Purchasing 185 is therefore a capacity stress, not a feasible age-compliant path. Its energy-table scope covers 85.55% of profile revenue miles.[46, 47, 48, 49]

London starts from 8,643 contracted positions, including 950 electric and 20 hydrogen. A 17-year cohort scenario renews 7,759 slots, converting 7,693 non-electric positions, with a peak of 776 slots/year; 14- and 12-year peaks are 977 and 2,568. Hydrogen stays outside the electric count until its source is certified. London lacks route blocks, depot power and local prices, so its financial and environmental application is unavailable.[50]

The later 2025 Investment Plan reports 4.424 million bus service hours in 2033, below the independently reported 2023 actual total. Because those accounting boundaries differ, we retain the 2023 service schedule as the hard floor rather than credit the transition with reduced operation; reconciling the later plan is a separate reporting task.[17, 14]

6.1 A solved age-cohort planning benchmark

For London, an original *fleet slot* denotes one available vehicle position of the same body type. Renewing a slot requires suitable electric capacity; the operational model may require additional vehicles. Let n_c be the observed size and b_c the registration year of cohort c, and choose a maximum chassis-life scenario L. These caps are assumptions for sensitivity, not asserted TfL rules. A non-electric cohort must renew by $d_c = \min(2033, \max(2024, b_c + L))$. An existing electric cohort enters the renewal model only when $b_c + L \leq 2033$; the 2013 Routemaster retains its chassis

Year	Age-due slots	Cumulative buys	Newly missed	Overdue after year
2024	539	185	354	354
2025	517	185	332	686
2026	0	0	0	501
2027	0	0	0	316
2028	0	0	0	131
2029	30	30	0	0
2030	63	63	0	0
2031	97	97	0	0
2032	177	177	0	0
2033	146	146	0	0

Table 10: Chicago 17-year hard-age stress under the unverified 185-bus/year capacity. The 539 vehicles in the oldest bins exceed 2024 capacity by 354 immediately. Clearing later backlog cannot undo that missed deadline. Bins are reported age intervals, so oldest-age deadlines are a robust stress rather than a vehicle register.

vintage despite its 2022 electric retrofit.[50]

Let integer a_{cy} allocate original slots to renewal years $2024 \le y \le d_c$. We first minimize the required peak annual renewal capacity M:

$$\min M : \qquad \sum_{y \le d_c} a_{cy} = n_c, \quad \sum_{c} a_{cy} \le M, \quad 0 \le a_{cy} \le n_c, \quad a_{cy} \in \mathbb{Z}_{\ge 0}. \tag{17}$$

With M fixed, a second solve minimizes $\sum_{c,y}(d_c - y)a_{cy}$, avoiding avoidable early renewal. Among ties, a third solve prioritizes earlier non-electric conversion without changing either optimum. This is an executed integer program using the public cohort counts, not an assumed uniform purchase schedule.

An independent certificate follows from $D_t = \sum_{c:d_c \le t} n_c$:

$$M^\star = \max_{t=2024,\ldots,2033} \left\lceil \frac{D_t}{t - 2023} \right\rceil. \tag{18}$$

Necessity follows because all D_t slots must fit into $t - 2023$ annual capacities. Sufficiency follows by allocating the earliest-deadline cohorts first when every cohort may renew from 2024. Later technical release dates, class-specific production limits and charging construction can increase the requirement; the certificate concerns this cohort submodel.

Life cap	Peak slots/year	Existing EV renewals	Total slots
12	2,568	735	8,428
14	977	247	7,940
17	776	66	7,759

Table 11: Optimal London fixed-availability cohort benchmarks, 2024–2033. Life caps are scenarios. Counts are renewed original fleet slots; operational feasibility can require additional buses.

All three scenarios were solved on CPU using SciPy's HiGHS mixed-integer solver; every phase reaches zero gap and the peak optima match the analytic bound. With a 17-year cap, 7,693

non-electric slots convert and 66 existing electric slots renew, requiring a peak of 776 original slots/year. The 14-year case needs 977 and renews 247 electric slots. The 12-year case creates a 2,568-slot catch-up in 2024. These are cohort-capacity results, not actual operating bus counts; facility release dates would invalidate the analytic sufficiency.

The 17-year scenario advances 21,908 slot-years relative to the assumed chassis-life deadlines, compared with 334 under the 14-year scenario. These are remaining-service exposures, not monetary losses: resale and ordinary-renewal cash flows are needed to value them. The benchmark provides a numerical ten-year procurement-capacity test while retaining the operational feasibility gate.

Year	Converted slots	EV renewals	Non-electric slots left
2024	775	0	6,918
2025	776	0	6,142
2026	776	0	5,366
2027	776	0	4,590
2028	776	0	3,814
2029	776	0	3,038
2030	775	1	2,263
2031	770	6	1,493
2032	769	7	724
2033	724	52	0

Table 12: An optimal London 17-year-cap fleet-slot roadmap. Like-body available slots are maintained; this table does not assign routes or chargers.

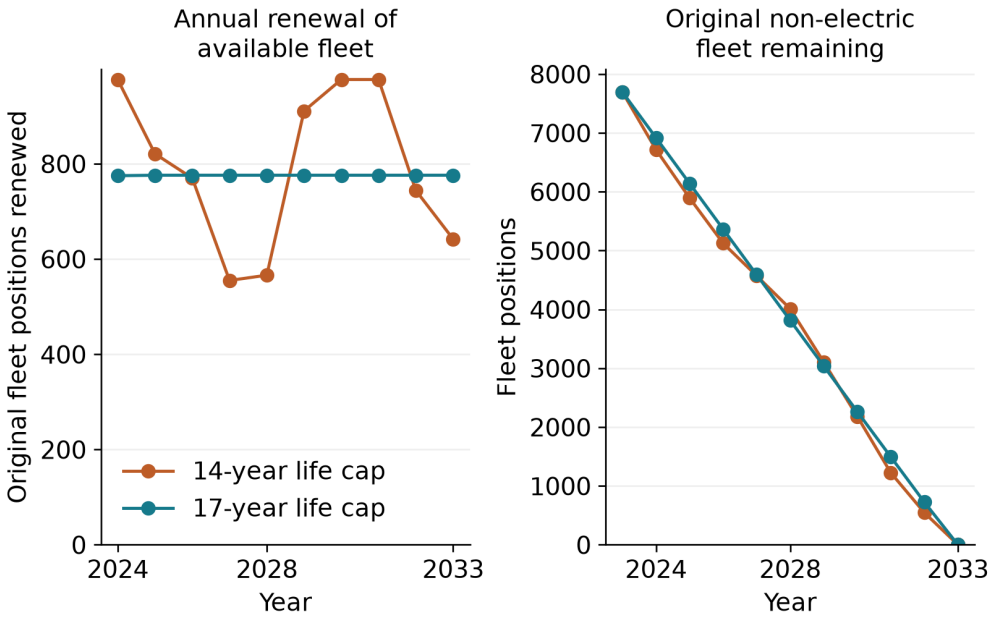


Figure 4: Executed London age-cohort roadmaps under fixed availability. The displayed numbers require compatible electric capacity, all-electric accessories, adequate chargers and trip checks before they become actual bus-purchase quantities.

7 Sensitivity, limitations and decision boundary

Reliable range is the dominant operational variable. At 149/150 km the weekday requires 2,444 sequential requests versus 1,502 participating blocks; at 300 km it requires 1,698–1,741, and at 350 km 1,587–1,623 because long duties remain. All 25 patterns at 300 km reach an upper-distance maximum of 1,745. A 745-km highway block requires terminal charging or exchange. These are distance screens; task timing and vehicle reuse can move the count in both directions.

The 432,768-cell grid varies the 1,711–1,733 stock scenario, seven paired ranges, 5–18 retained BEBs, 9/17–11/17 renewal, a 1.5–5.0% rate, vehicle cost basis and four fuel intensities. The annual service kilometre denominator is held fixed; fleet variation changes capital and position counts alone. The named cell reproduces CAD 4.303 billion and −89.2% exactly. Median gross PV is CAD 4.78 billion; economic median and 5th–95th are −114.0% and −140.2 to −83.4%; environment median and 5th–95th are 7.8% and −33.7 to 15.2%. These are equal-weight assumptions, not confidence intervals.

Funding does not alter resource cost. In the generic case, 0/25/50% grants change agency PV from CAD 4.30 to 3.40 and 2.50 billion, and the score from −89.2 to −49.5 and −9.8%. The class central case gives −68.8/ − 38.8/ − 5.5%. Common trolley renewal receives the same percentage in both comparator and plan; the former one-sided grant case is retained only as an explicit policy counterfactual. The transition still needs policy value or lower bids. The conventional O&M rate sensitivity remains conditional. The regional memo gives CAD 0.62–0.97 billion annual gross-value-added proxy and 6.0–8.8 thousand FTE for the wider Bus Operations boundary; construction-period output is separate and non-additive.[14, 41, 42]

The environmental sign is not robust. Generic scores are −17.6%, 8.9%, 12.4% and 15.1%; class scores are −14.3%, 10.5%, 13.6% and 16.2%. A stress replacement of the 323-bus first delivery cohort adds 6.276 kt and moves the generic renewable-fuel case to −19.3%. This is not a predicted replacement year. The 0.302–1.58 kg CO_2e/km baseline and 19 g/km electric rates remain transfer-sensitive, and depot construction/end-of-life are excluded.

The remaining uncertainties are a January–August timetable or as-operated records; block-to-vehicle and block-to-garage links; vehicle ages and payload; depot parking, ports, firm power and commissioning dates; terminal exchange times; trolley feeder capacity; and 2033 winter tests. Asset and infrastructure terminal values are now bracketed, but uniform initial ages and resale shares remain scenarios; future battery replacement lies beyond this purchase schedule and is not capitalized. No city-wide Pareto point or full TCO is certified.

The implementation gates are: every recovered scheduled trip is retained; every candidate day satisfies class, capacity, road transfer, winter SOC, charger/site and periodic closure; and every cost and emission term has a source, year, unit and uncertainty range. The validator returns `UNVERIFIED` when the second or third lacks inputs.

7.1 Conclusion

The study supports a conditional plan, not a 2033 procurement order. The 1,647 route vehicles plus 262 trolley renewals zero stock arithmetically; service preservation and 18 initial BEBs remain unproved. Central constant-price PV is CAD 4.43 versus 2.62 billion (−68.8%); asset/infrastructure and environment ranges are −41 to −106% and −14.3 to 16.2%. Commission grid and depot work, certify terminal/winter operation, and gate each post-2030 cohort on site, SOC and bid evidence.

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Letter to a Metro Vancouver decision-maker

To: TransLink Board and Mayors' Council
Subject: Conditions for a zero-tailpipe fixed-route bus fleet by 2033

Dear decision-makers,

We treat 2033 as an accelerated target against TransLink's 2040 goal. The recovered fall-2023 GTFS schedules 54,497 fixed-route trips across 25 patterns; the representative weekday has 1,752 source blocks and 1,087 simultaneous trips. We would retain the recovered departures, directions and capacity proxies. Existing trolleybuses remain in the electric baseline. Strict clean-electricity sourcing is unverified.

At 300 km, the weekday yields 1,698–1,741 distance requests for 1,502 participating blocks, while all 25 patterns reach at most 1,745. Route 100 supplies a small terminal-linked sample: 223 trips and 23 blocks become 25 duty fragments, with a simultaneous-trip lower bound of 14. A 700-kWh double-decker leaves 17 of 57 duties short of one overnight charge; a 150-kWh shuttle leaves 49 of 101. Garage, charger and payload records remain absent.

The assumed class inventory buys 1,647 route vehicles and 262 common trolley renewals. Constant 2023 gross PV is CAD 4.43 billion against CAD 2.62 billion (-68.8%). With common trolley renewal treated symmetrically, 25/50% grants give $-38.8/-5.5\%$. Retained-asset and infrastructure terminal-value cases span -41 to -106%. The 1,927-electric-asset endpoint is stock arithmetic, not a dispatch certificate.

The partial production/operation/maintenance boundary yields -14.3 to 16.2% in the class case. The 2023 tailpipe proxy is 126–408 t NO_x, 1.11–8.34 t $PM_{2.5}$ and 2.34–14.18 t PM_{10} per year. Depot construction and end-of-life remain excluded. The Bus Operations umbrella is associated with CAD 0.62–0.97 billion annual value-added and 6.0–8.8 thousand FTE; construction effects are separate.

Start depot design, interconnection and permits in 2024; no conversion is credited before 2030. Determine quantities through dispatch and commissioned-site checks, test 40- and 60-foot buses under all-electric winter conditions, and hold double-deck and shuttle conversion behind terminal charging and payload certification. If a duty fails time, SOC or site capacity, publish that gap rather than deleting service.

Sincerely,
Fleet Transition Study Team

Report on Use of AI

Tools and versions. OpenAI Codex (GPT-6) and deepseek-flash were used in later sessions for research planning, source discovery, code implementation, data audits, mathematical review, and English drafting. No original 2023 HiMCM solution paper was requested or used.[51]

Queries and work products. The prompts requested: (i) an evidence-locked plan for environmental, financial and three-city fleet-transition models; (ii) independent literature and primary-source discovery, explicitly excluding the 2023 HiMCM solution; (iii) implementation and testing of trip/block, reserve, charger and tariff models; (iv) recovery and audit of public 2023 TransLink GTFS; (v) a high-standard LaTeX manuscript and reproducible repository; and (vi) completion of missing inputs without invented provenance; and (vii) closure of an independent adversarial review with regenerated evidence, tests and manuscript. The corresponding outputs are retained in `src/`, `data/processed/`, `results/`, this manuscript, and the GitHub history. `research/ai_query_log.md` retains the original-language request texts and their artifact mapping; `research/ai_use_report.md` records the broader source and verification boundary.

Verification. Source URLs, dates, pages, units and hashes are retained in the parameter ledger and provenance files. GTFS files match immutable Git blobs. The model suite includes explicit tests for schedule time, 2033 electrification, energy reserve, charger concurrency, tariff bounds, cost transfer and cohort feasibility. Computed values in this paper are traceable to named scripts. The human team must still verify every cited fact and decide what to submit.

Request and output index

The original Chinese request texts are retained without paraphrasing in `research/ai_query_log.md`. The following is the working English summary of each request and its complete retained artifact set; internal reasoning traces are not treated as research outputs.

1. *Plan the study, choose a well-documented metropolitan case with fallback cities, and search independent electric-bus/fleet/infrastructure literature without using the 2023 HiMCM solution.* Outputs: research boundary, literature notes, parameter ledger, and the three-city design.

2. *Use LaTeX, research other-year award-winning paper presentation, standardize the format, and run the calculations on the available CPU.* Outputs: the canonical TeX/PDF, GTFS and schedule audits, charging LP, London MILP, 432,768-case grid, scenario results, figures, and format checks.

3. *Fill missing fleet, infrastructure, double-deck/shuttle, regional economic and air-quality evidence while refusing to invent unavailable records.* Outputs: APTA negative-evidence audit, deck/shuttle technology audit, regional/NOx/PM audit, class-specific cost bound, updated ledger and manuscript.

4. *Verify all figures and tests, preserve reproducibility, and synchronize the project to the authorized private GitHub repository when access allows.* Outputs: the local Git history, `scripts/reproduce_local.py`, manuscript-number and PDF checks, and the repository-ready file set.